Conditional Canonical Lifts and a Direct Information-Theoretic Tomography Bound

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Abstract

We give a direct information-theoretic proof of a nearly sample-optimal upper bound for quantum state tomography in infidelity. The central statement is a conditional canonical-lift inequality. Given a classical label X, quantum side information B, and a state ρ_x on a d-dimensional register C, let H hold the canonical purification $|\sqrt{\rho_x}\rangle\!\rangle$. We prove

$$I(X : H \mid B) \le \Phi_{d^2}\left(1 - e^{-2I(X:C|B)}\right),$$

where $\Phi_D(t) = h_2(t) + t\log(D-1)$ up to its maximum $\log D$. The main ingredient is a pretty-good-measurement affinity bound proved by Schatten-norm interpolation. Combining this inequality with measured-relative-entropy recovery and Schur–Weyl compression yields, for rank at most r,

$$N = O\left(\frac{rd}{\varepsilon}\log^2\frac{ed}{\varepsilon}\right)$$

copies for constant-success tomography to infidelity ε. The proof treats mixed states directly: it uses neither a reduction to pure-state tomography nor a random-purification channel.

1 Introduction

Let ρ be an unknown density operator on $\mathbb{C}^d$. We use squared Uhlmann fidelity

$$F(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2$$

and measure tomography error by the infidelity $1 - F(\rho,\widehat{\rho})$. Assume $d \ge 2$, the case $d = 1$ being trivial. Yuen proved the information-theoretic lower bound $\Omega(rd/\varepsilon)$ for rank-r states [7], while Haah, Harrow, Ji, Wu, and Yu gave the standard collective-measurement upper bound $\widetilde{O}(rd/\varepsilon)$ [3].

This note gives a different, direct proof of the upper bound, losing only two logarithmic factors. The argument is an extension of the conditional-mutual-information and recovery-map method used for Best Observable Selection: one identifies a copy whose predictive information is small, applies a recovery theorem, and evaluates the recovered classical decision by a conceptual holdout test. The new issue is that mixed-state fidelity is nonlinear in the unknown state, so a one-copy observable cannot serve as the holdout test. The replacement is the *canonical purification*.

Fix a basis and vectorize operators in that basis. For a density operator ρ, write

$$|\gamma_\rho\rangle := |\sqrt{\rho}\rangle\!\rangle \in \mathbb{C}^d \otimes \mathbb{C}^d, \qquad \gamma_\rho := |\gamma_\rho\rangle\langle\gamma_\rho|.$$

This is a unit vector, and

$$\langle \gamma_\sigma \mid \gamma_\rho \rangle = \operatorname{Tr} \sqrt{\sigma}\sqrt{\rho} =: a(\rho, \sigma).$$

We call $a(\rho, \sigma)$ the *canonical affinity*. Since

$$a(\rho, \sigma) \leq \|\sqrt{\rho}\sqrt{\sigma}\|_1,$$

we have

$$1 - F(\rho, \sigma) \leq 1 - a(\rho, \sigma)^2. \tag{1}$$

The core theorem says that, even in the presence of noncommuting quantum side information, little information about one fresh copy implies little information about the canonical purification.

For $D \geq 2$, define the capped Fano function

$$\Phi_D(t) := \begin{cases} h_2(t) + t \log(D-1), & 0 \leq t \leq 1 - 1/D, \\ \log D, & 1 - 1/D < t \leq 1, \end{cases} \tag{2}$$

where $h_2(t) = -t \log t - (1-t)\log(1-t)$ and all logarithms are natural. This function is increasing and concave.

Theorem 1.1 (Conditional canonical lift)**.** *Let*

$$\omega_{XBC} = \sum_x p_x |x\rangle\langle x|_X \otimes \beta_x^B \otimes \rho_x^C,$$

where $C \cong \mathbb{C}^d$, and let

$$\omega_{XBH}^\sharp = \sum_x p_x |x\rangle\langle x|_X \otimes \beta_x^B \otimes \gamma_{\rho_x}^H, \qquad H \cong \mathbb{C}^{d^2}.$$

Then

$$I(X : H \mid B)_{\omega^\sharp} \leq \Phi_{d^2}\left(1 - e^{-2I(X:C|B)_\omega}\right). \tag{3}$$

In particular, when $I(X : C \mid B)_\omega \leq \frac{1}{2}$,

$$I(X : H \mid B)_{\omega^\sharp} = O\left(I(X : C \mid B)_\omega \log \frac{ed^2}{I(X : C \mid B)_\omega}\right).$$

The proof has a short conceptual outline. First measure B with the pretty-good measurement for the ensemble carried by B. Given outcome y, use γ_{ρ_y} as a candidate prediction for H. A Fano bound reduces the theorem to showing that the PGM's average canonical affinity is at least $e^{-I(X:C|B)}$. The latter statement is proved by an interpolation family whose order-1 endpoint is the conditional mutual information and whose order-2 endpoint is the PGM affinity.

The tomography consequence can be stated exactly in terms of the Schur–Weyl compression dimension. Let $\mathcal{Q}_\lambda^{(d)}$ denote the irreducible $U(d)$ representation indexed by a partition $\lambda \vdash N$, and set

$$K_{N,d,r} := \sum_{\substack{\lambda \vdash N \\ \ell(\lambda) \leq r}} \dim \mathcal{Q}_\lambda^{(d)}. \tag{4}$$

Theorem 1.2 (Direct tomography bound)**.** *For every $1 \le r \le d$ and $N \ge 1$, there is a collective measurement on N copies, followed by a classical state-valued estimator $\widehat{\rho}$, such that*

$$\sup_{\mathrm{rank}(\rho)\le r} \mathbb{E}\big[1 - F(\rho, \widehat{\rho})\big] \le \Phi_{d^2}\Big(1 - K_{N,d,r}^{-2/N}\Big). \tag{5}$$

Moreover,

$$K_{N,d,r} \le (N+1)^{r-1}(N+d)^{r(2d-r-1)/2}, \tag{6}$$

and hence

$$\sup_{\mathrm{rank}(\rho)\le r} \mathbb{E}\big[1 - F(\rho, \widehat{\rho})\big] = O\left(\min\left\{1, \frac{rd\,\log(N+d)\,\log(edN)}{N}\right\}\right). \tag{7}$$

Consequently, $O((rd/\varepsilon)\log^2(ed/\varepsilon))$ copies suffice for expected infidelity at most ε. With the same asymptotic number of copies, the infidelity is at most ε with probability at least $2/3$.

The estimator is information-theoretic. No computational efficiency is claimed.

2 Preliminaries

For a state τ, let $S(\tau) = -\operatorname{Tr}(\tau \log \tau)$. We use the standard notation $I(A : C \mid B)$ for quantum conditional mutual information. The measured relative entropy is

$$D_{\mathrm{meas}}(\rho\|\sigma) := \sup_{\mathcal{M}} D_{\mathrm{KL}}(\mathcal{M}(\rho)\|\mathcal{M}(\sigma)),$$

where the supremum ranges over quantum-to-classical channels.

We use the following strengthening of the Fawzi–Renner recovery theorem [2], due to Brandão, Harrow, Oppenheim, and Strelchuk [1].

Theorem 2.1 (Measured-relative-entropy recovery)**.** *For every tripartite state τ_{ABC}, there is a channel $\mathcal{R}_{B\to BC}$ such that*

$$D_{\mathrm{meas}}(\tau_{ABC}\|(\mathrm{id}_A \otimes \mathcal{R}_{B\to BC})(\tau_{AB})) \le I(A : C \mid B)_\tau.$$

We will also use the elementary entropy bound behind Fano's inequality.

Lemma 2.2 (One large eigenvalue)**.** *Let τ be a state on a D-dimensional space. If some unit vector $|v\rangle$ satisfies $\langle v|\tau|v\rangle \ge 1 - t$, then*

$$S(\tau) \le \Phi_D(t).$$

Proof. The largest eigenvalue of τ is at least $1-t$. Subject to that constraint, entropy is maximized by placing the remaining mass uniformly on the other $D-1$ eigenvalues. This gives the first branch of (2); when that expression has passed its maximum, the trivial bound $S(\tau) \le \log D$ is better. □

3 The conditional canonical-lift inequality

We now prove Theorem 1.1. Throughout this section, set

$$u := I(X : C \mid B)_\omega, \qquad v := I(X : H \mid B)_{\omega^\sharp}, \qquad D := d^2.$$

3.1 The PGM reduction

We may first assume that every β_x is pure. Indeed, choose a pure-state decomposition $\beta_x = \sum_z r(z \mid x)|b_{xz}\rangle\langle b_{xz}|$ and refine the classical label from x to (x, z). Conditioned on $X = x$, the register C is product with ZB, and the same is true of H. The chain rule therefore gives

$$I(XZ : C \mid B) = I(X : C \mid B), \qquad I(XZ : H \mid B) = I(X : H \mid B).$$

Relabeling (x, z) as x, write $\beta_x = |b_x\rangle\langle b_x|$.

Let

$$\overline{\beta} := \sum_x p_x |b_x\rangle\langle b_x|.$$

On $\mathrm{supp}(\overline{\beta})$, the pretty-good measurement is

$$E_y := p_y\, \overline{\beta}^{-1/2} |b_y\rangle\langle b_y| \overline{\beta}^{-1/2}. \tag{8}$$

Its input-output joint distribution is

$$q_{xy} := p_x \langle b_x | E_y | b_x \rangle = p_x p_y \left| \langle b_x | \overline{\beta}^{-1/2} | b_y \rangle \right|^2. \tag{9}$$

Thus $q_{xy} = q_{yx}$, and both marginals of q are p.

Set

$$a_{xy} := a(\rho_x, \rho_y) = \mathrm{Tr}\, \sqrt{\rho_x}\sqrt{\rho_y} = \langle \gamma_{\rho_y} \mid \gamma_{\rho_x} \rangle. \tag{10}$$

The matrix (a_{xy}) is a correlation matrix, and $0 \le a_{xy} \le 1$.

Measure B with the PGM and call the outcome Y. Since H is pure conditioned on X,

$$v = I(X : H \mid B) = S(H \mid B).$$

Applying a channel to the conditioning system can only increase conditional entropy, so

$$S(H \mid B) \le S(H \mid Y).$$

Conditioned on $Y = y$, the state of H is

$$\tau_y = \sum_x \frac{q_{xy}}{p_y} \gamma_{\rho_x}.$$

Its overlap with the candidate γ_{ρ_y} is

$$\langle \gamma_{\rho_y} | \tau_y | \gamma_{\rho_y} \rangle = \sum_x \frac{q_{xy}}{p_y} a_{xy}^2.$$

By Lemma 2.2 and concavity of Φ_D,

$$\begin{aligned} v &\le \sum_y p_y S(\tau_y) \\ &\le \Phi_D\left(1 - \sum_{x,y} q_{xy} a_{xy}^2\right). \end{aligned} \tag{11}$$

Define the PGM affinity score

$$q_* := \sum_{x,y} q_{xy} a_{xy}. \tag{12}$$

Jensen's inequality gives

$$\sum_{x,y} q_{xy} a_{xy}^2 \geq q_*^2. \tag{13}$$

It remains to lower-bound q_*.

Lemma 3.1 (PGM affinity bound)**.** *In the setting above,*

$$q_* \geq e^{-u}. \tag{14}$$

The proof is the main technical step.

3.2 A Rényi bridge from conditional information to PGM affinity

Proof of Lemma 3.1. Let

$$M := \sum_x p_x |b_x\rangle\langle b_x| \otimes \rho_x.$$

Because the B-states are pure,

$$u = S(M) - S(\overline{\beta}) - \sum_x p_x S(\rho_x). \tag{15}$$

For $1 \leq \alpha \leq 2$, define

$$N_\alpha := \sum_x p_x |b_x\rangle\langle b_x| \otimes \rho_x^{1/\alpha}, \tag{16}$$

$$Q_\alpha := \mathrm{Tr}\left[(\overline{\beta}^{\frac{1-\alpha}{2\alpha}} \otimes \mathbb{1}) N_\alpha (\overline{\beta}^{\frac{1-\alpha}{2\alpha}} \otimes \mathbb{1}) \right]^\alpha. \tag{17}$$

All inverses and powers are taken on supports. We first record the endpoints:

$$Q_1 = 1, \qquad Q_2 = q_*. \tag{18}$$

The first identity is immediate. Expanding the square at $\alpha = 2$ gives

$$\begin{aligned} Q_2 &= \sum_{x,y} p_x p_y \left| \langle b_x | \overline{\beta}^{-1/2} | b_y \rangle \right|^2 \mathrm{Tr}\, \sqrt{\rho_x}\sqrt{\rho_y} \\ &= q_*. \end{aligned}$$

We next prove the interpolation inequality

$$Q_\alpha \leq Q_2^{\alpha - 1} \qquad (1 \leq \alpha \leq 2). \tag{19}$$

Introduce

$$\mathcal{V} := \sum_x \sqrt{p_x}\, |b_x\rangle\langle x| \otimes \mathbb{1}_C, \qquad \Sigma := \mathcal{V}\mathcal{V}^\dagger = \overline{\beta} \otimes \mathbb{1}_C, \qquad \Delta := \sum_x |x\rangle\langle x| \otimes \rho_x.$$

Then $N_\alpha = \mathcal{V} \Delta^{1/\alpha} \mathcal{V}^\dagger$. Write the polar decomposition

$$\mathcal{V} = \Sigma^{1/2} U,$$

where U is a partial isometry. A direct factorization of (17) yields

$$Q_\alpha^{1/(2\alpha)} = \| \Sigma^{1/(2\alpha)} U \Delta^{1/(2\alpha)} \|_{2\alpha}. \tag{20}$$

Consider the operator-valued analytic function on the strip $0 \le \Re z \le 1$,

$$F(z) := \Sigma^{1/2-z/4} U \Delta^{1/2-z/4}. \tag{21}$$

On the two boundary lines, unitary invariance of Schatten norms gives

$$\sup_t \|F(it)\|_2 = \|\Sigma^{1/2} U \Delta^{1/2}\|_2 = \|\mathcal{V}\Delta^{1/2}\|_2 = 1, \tag{22}$$

$$\sup_t \|F(1+it)\|_4 = \|\Sigma^{1/4} U \Delta^{1/4}\|_4 = Q_2^{1/4}. \tag{23}$$

For

$$\theta := \frac{2(\alpha-1)}{\alpha},$$

we have

$$\frac{1}{2\alpha} = \frac{1-\theta}{2} + \frac{\theta}{4}, \qquad \frac{1}{2} - \frac{\theta}{4} = \frac{1}{2\alpha}.$$

The Schatten-space three-lines theorem (see, e.g., [6]) therefore implies

$$\|\Sigma^{1/(2\alpha)} U \Delta^{1/(2\alpha)}\|_{2\alpha} \le Q_2^{\theta/4}.$$

Raising to the power 2α proves (19).

Finally, differentiate at the order-1 endpoint. Since $Q_1 = 1$, direct differentiation of (17) gives

$$\begin{aligned} \frac{d}{d\alpha} \log Q_\alpha \Big|_{\alpha=1} &= \operatorname{Tr}(M \log M) - \operatorname{Tr}(\overline{\beta} \log \overline{\beta}) - \sum_x p_x \operatorname{Tr}(\rho_x \log \rho_x) \\ &= -u, \end{aligned} \tag{24}$$

where the last equality is (15). Taking logarithms in (19), dividing by $\alpha - 1$, and sending $\alpha \downarrow 1$ yields

$$-u \le \log Q_2 = \log q_*.$$

Equivalently, $q_* \ge e^{-u}$.

The argument above was written for strictly positive operators. Replacing every ρ_x by $(1-\delta)\rho_x + \delta \mathbb{1}/d$ and then sending $\delta \downarrow 0$ gives the general case; $\overline{\beta}$ is simply restricted to its support. □

3.3 Completion of the lift theorem

Proof of Theorem 1.1. Combining (11), (13), and Lemma 3.1,

$$v \le \Phi_D(1 - q_*^2) \le \Phi_D(1 - e^{-2u}),$$

which is (3).

For the small-u estimate, use $1 - e^{-2u} \le 2u$ and

$$\Phi_D(t) \le t \log \frac{eD}{t}$$

throughout the increasing branch of Φ_D; outside that branch the claimed big-O estimate is trivial after adjusting the universal constant. □

4 From the lift theorem to tomography

We now prove Theorem 1.2. The proof is first Bayesian; minimax then gives a single worst-case estimator.

4.1 Small predictive information gives a state estimator

Fix a finite prior $\{p_x, \rho_x\}$ on states of rank at most r. Let $B_1, \ldots, B_N$ hold conditionally independent copies of ρ_x, and let H hold the canonical purification. Thus

$$\Omega_{XB^NH} = \sum_x p_x |x\rangle\langle x|_X \otimes \rho_x^{\otimes N} \otimes \gamma_{\rho_x}^H. \tag{25}$$

For $0 \le t < N$, set

$$\eta_t := I(X : B_{t+1} \mid B_1 \cdots B_t)_\Omega.$$

Suppose for the moment that $\eta_t \le \eta$. Applying Theorem 1.1 with $B = B_1 \cdots B_t$ and $C = B_{t+1}$ gives

$$\zeta_t := I(X : H \mid B_1 \cdots B_t) \le \Phi_{d^2}(1 - e^{-2\eta}). \tag{26}$$

By Theorem 2.1, with the roles of the two outer registers interchanged, there is a channel

$$\mathcal{R} : B_1 \cdots B_t \longrightarrow \widehat{X} B_1 \cdots B_t$$

such that, after dephasing $\widehat{X}$ in the label basis,

$$D_{\mathrm{meas}}\Big(\Omega_{XB^tH} \Big\| \widetilde{\Omega}_{\widehat{X}B^tH}\Big) \le \zeta_t. \tag{27}$$

The Bayesian estimator applies this channel, measures $\widehat{X}$, and outputs $\rho_{\widehat{X}}$.

For analysis, measure XH or $\widehat{X}H$ with the controlled effect

$$T := \sum_x |x\rangle\langle x| \otimes \gamma_{\rho_x}. \tag{28}$$

On the ideal state this test accepts with probability 1. On the recovered state it accepts with probability

$$q = \mathbb{E}_{x,\widehat{x}} \left[a(\rho_x, \rho_{\widehat{x}})^2 \right].$$

Data processing in (27) therefore gives

$$-\log q = D_{\mathrm{KL}}(\mathrm{Bern}(1) \| \, \mathrm{Bern}(q)) \le \zeta_t.$$

Using (1),

$$\mathbb{E}[1 - F(\rho_X, \rho_{\widehat{X}})] \le 1 - q \le 1 - e^{-\zeta_t} \le \zeta_t. \tag{29}$$

Thus the only remaining task is to find a t with small η_t.

4.2 The Schur–Weyl information budget

Under Schur–Weyl duality [4],

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\substack{\lambda \vdash N \\ \ell(\lambda) \le d}} \mathcal{Q}_\lambda^{(d)} \otimes \mathcal{P}_\lambda, \tag{30}$$

where $\mathcal{P}_\lambda$ is the corresponding irreducible symmetric-group representation. Every tensor power $\rho^{\otimes N}$ commutes with the symmetric-group action and hence has the form

$$\rho^{\otimes N} = \bigoplus_\lambda A_\lambda(\rho) \otimes \frac{\mathbb{1}_{\mathcal{P}_\lambda}}{\dim \mathcal{P}_\lambda}. \tag{31}$$

If $\operatorname{rank}(\rho) \le r$, only partitions with $\ell(\lambda) \le r$ occur. Consequently, the multiplicity registers $\mathcal{P}_\lambda$ contain no information about ρ and can be discarded reversibly on this family. The informative compressed Hilbert space has dimension $K_{N,d,r}$ from (4). Therefore

$$I(X : B_1 \cdots B_N)_\Omega \le \log K_{N,d,r}. \tag{32}$$

By the chain rule,

$$\sum_{t=0}^{N-1} \eta_t = I(X : B_1 \cdots B_N)_\Omega.$$

Hence some $t < N$ satisfies

$$\eta_t \le \frac{\log K_{N,d,r}}{N}. \tag{33}$$

For completeness, the Weyl dimension formula gives the elementary bound in (6). There are at most $(N+1)^{r-1}$ partitions of N with at most r nonzero parts. For such a partition,

$$\dim \mathcal{Q}_\lambda^{(d)} = \prod_{1 \le i < j \le d} \frac{\lambda_i - \lambda_j + j - i}{j - i}.$$

The factors with $i > r$ equal 1, and there are

$$\sum_{i=1}^{r} (d - i) = \frac{r(2d - r - 1)}{2}$$

remaining factors, each at most $N + d$. This proves (6), and in particular

$$\log K_{N,d,r} \le rd \log(N + d). \tag{34}$$

4.3 Proof of the tomography theorem

Proof of Theorem 1.2. For the fixed finite prior, choose t as in (33). Equations (26) and (29) give a Bayesian estimator with risk at most

$$\Phi_{d^2}\left(1 - \exp\left[-\frac{2 \log K_{N,d,r}}{N}\right]\right) = \Phi_{d^2}\left(1 - K_{N,d,r}^{-2/N}\right).$$

This bound is independent of the prior. The standard compact minimax theorem for quantum statistical decision problems [5] therefore gives a single POVM estimator with the same worst-case

bound. Equivalently, one may first apply finite minimax on increasingly fine finite state sets and then pass to a limit using compactness and continuity of fidelity. This proves (5).

To derive (7), let

$$\eta := \frac{\log K_{N,d,r}}{N}.$$

When η is bounded below by a universal constant, the trivial risk bound 1 suffices. Otherwise,

$$\Phi_{d^2}(1 - e^{-2\eta}) \le O\left(\eta \log \frac{ed^2}{\eta}\right).$$

Using (34) and $r \le d$ gives

$$\eta \log \frac{ed^2}{\eta} = O\left(\frac{rd \, \log(N+d) \, \log(edN)}{N}\right),$$

which is (7).

Finally, taking

$$N = C\frac{rd}{\varepsilon} \log^2 \frac{ed}{\varepsilon}$$

for a sufficiently large universal constant C makes the expected infidelity at most ε. Markov's inequality converts expected infidelity $\varepsilon/3$ into infidelity at most ε with probability at least $2/3$, at only a constant-factor increase in N. □

5 Remarks

Where the logarithms enter. One logarithm comes from the Schur–Weyl information budget $\log K_{N,d,r}$; the second comes from lifting a fresh mixed state to its d^2-dimensional canonical purification through the Fano function. Removing either logarithm would require a sharper ingredient than the present proof, not merely a tighter solution of the final sample-complexity inequality.

Directness. The canonical purification is used only as a conceptual holdout register in the analysis. The algorithm never receives, constructs, or measures a purification of the unknown state. The physical procedure acts solely on the original copies through the recovery channel supplied by Theorem 2.1.

Computational status. The proof is existential in two places: the measured-relative-entropy recovery map and the minimax passage. It therefore gives an information-theoretic measurement, not an efficient tomography algorithm.

Relation to Best Observable Selection. The logical pattern is the same as in the short BOS proof: a chain rule exposes a low-CMI time, recovery predicts a classical decision from the preceding copies, and a controlled holdout test turns measured relative entropy into loss. The conditional canonical-lift theorem is the new step that makes this pattern work for nonlinear mixed-state fidelity.

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